

## GENERALIZED NON-LINEAR $\sigma$ -MODELS WITH GAUGE INVARIANCE

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Generalizations of the non-linear  $\sigma$ -model and of the  $CP^N$  model are considered. They are invariant under a global group  $G$  and under a local gauge subgroup  $H$ . Whenever  $G/H$  defines a symmetric space, we can show that these models are asymptotically free. Furthermore, in two dimensions the local gauge invariance leads to confinement.

### 1. Introduction

The asymptotic freedom of two-dimensional non-linear  $\sigma$ -models, the existence of an infinite number of classical local or non-local conservation laws, the non-perturbative character of the particle spectrum are some of the features which explain the current interest in these models. Generalizations of the usual  $O(N)$  invariant models have been proposed which in addition to a global symmetry have a local gauge invariance [1, 2]. The case corresponding to an abelian symmetry, has been extensively studied recently: it is called the  $CP^N$  model [3, 4]. In two dimensions, at least in the large  $N$  limit, it is known that the constituent particles are confined into bound multiplets by long range forces [4]. We have investigated the models corresponding to symmetric spaces which provide the largest class that we know which has an infinite number of classical non-local conservation laws and a single coupling constant even in the quantized theory. Asymptotic freedom of these models is established. In addition for some of them which are non-abelian generalizations of the  $CP^N$  model we can show that the same confinement mechanism takes place at least in the large- $N$  limit.

In sect. 2 we recall how these models are constructed, and why they have the classical equations of motion which are sufficient to prove the existence of an infinite number of conservation laws. Then quantization is considered for the models  $U(N)/U(p) \times U(N-p)$  and for their orthogonal analogue. Two-loop results for the renormalization group functions are given. In sect. 5 we prove that the

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general symmetric spaces models are asymptotically free. We end with the large- $N$  limit.

## 2. Generalized $\sigma$ -models

Several recent investigations [1, 2] have considered generalizations of the non-linear  $\sigma$ -model which are of interest in particular since they all possess classically an infinite number of non-local conservation laws in two dimensions [2]. These models derive from an euclidean action

$$\begin{aligned} A &= \frac{1}{2} \int d^2x \operatorname{tr} [\partial_\mu g(x) \partial_\mu g^{-1}(x)] \\ &= -\frac{1}{2} \int d^2x \operatorname{tr} (g^{-1} \partial_\mu g)(g^{-1} \partial_\mu g), \end{aligned} \quad (1)$$

in which  $g(x)$  belongs to some compact simple Lie group in a matrix representation\*. For non-simple groups the action reduces to a sum of independent terms corresponding to the simple components. Different models correspond to  $g(x)$  varying over various subsets of the whole group  $G$ . It has been shown by Eichenherr and Forger [6] that the cases for which an infinite number of non-local conservation laws can be constructed correspond to a geometric structure of symmetric spaces. When the group  $G$  is either the unitary or the orthogonal group the corresponding models have been described by Pisarski [6]. We shall list them again here and show directly that they lead to classical conservation laws by the method used in ref. [2]. There it was shown that the conditions could be expressed on the equations of motion. An explicit construction was given provided the classical equations of motion could be expressed as

$$\partial_\mu J_\mu = 0, \quad (2a)$$

$$J_\mu = g^{-1}(x) \partial_\mu g(x). \quad (2b)$$

We will list the models for which these conditions are satisfied.

### 2.1. Chiral models

The matrix  $g(x)$  varies over the whole group  $G$ . The action is invariant under the global transformations of  $G \times G$

$$g(x) \rightarrow h_1 g(x) h_2.$$

\* Non-linear realizations of symmetry groups in quantum field theory have been extensively discussed in the past. In particular the article by Coleman, Wess and Zumino [5] contains several discussions relevant to our formalism.

The classical equations of motion are obtained from the variation

$$\begin{aligned} g(x) &\rightarrow g(x)(1 + \epsilon(x)), \\ J_\mu(x) &= g^{-1} \partial_\mu g \rightarrow J_\mu + \partial_\mu \epsilon + [J_\mu, \epsilon], \\ A &\rightarrow A - \int dx \operatorname{Tr} \{J_\mu (\partial_\mu \epsilon + [J_\mu, \epsilon])\} = A + \int dx \operatorname{Tr} [(\partial_\mu J_\mu) \epsilon]. \end{aligned}$$

Consequently since this variation should vanish for any  $\epsilon$  in the Lie algebra of  $G$ , the classical equations of motion express the conservation of the current  $J_\mu$ :  $\partial_\mu J_\mu = 0$  and equations (2) are satisfied.

## 2.2. Second class

The matrices  $g(x)$  vary over a subset  $F$  of the group  $G$  defined in the following way. Let  $g_0$  be a given fixed element of  $G$  which satisfies the condition

$$g_0 g_0^* = \lambda \mathbb{1}, \quad (3)$$

in which  $\lambda$  is a real or complex number. The star operation is an involutive automorphism of the group, i.e.  $(g_1 g_2)^* = g_1^* g_2^*$ ,  $(g^*)^* = g$ . This automorphism can be trivial  $g^* = g$ . The subset  $F$  will be defined to be the set of matrices  $g(x)$  of the form

$$g(x) = u^{-1*}(x) g_0 u(x) \quad (4)$$

in which  $u$  is an arbitrary element of  $G$ ; from these definitions it follows immediately that

$$g(x) g^*(x) = \lambda \mathbb{1}. \quad (5)$$

Let  $H$  be the subgroup of elements  $h(x)$  which leave  $g_0$  invariant:

$$g_0 = h^{-1*} g_0 h \Leftrightarrow h \in H. \quad (6)$$

The elements  $g(x)$  are invariant under a left multiplication of  $u(x)$  by an element of  $H$  and they belong to the (left) coset space  $G/H$ . Eq. (6) may also be written in the form  $g_0^{-1} h^* g_0 = h$  which shows that  $H$  is the subgroup of  $G$  invariant under the involutive automorphism  $u \rightarrow \bar{u} = g_0^{-1} u^* g_0$ . This is the definition of a symmetric space  $G/H$  corresponding to the automorphism  $u \rightarrow \bar{u}$ . The action on  $G/H$ , again defined as

$$A = \frac{1}{2} \int dx \operatorname{tr} \partial_\mu g(x) \partial_\mu g^{-1}(x),$$

leads to equations of motion of the desired form [2]. Indeed let us consider an arbitrary variation

$$u(x) \rightarrow u(x)(1 + \epsilon(x)),$$

and correspondingly

$$g(x) \rightarrow (1 - \epsilon^*)g(1 + \epsilon),$$

$$A \rightarrow A + \int dx \operatorname{tr} [\epsilon \partial_\mu (g^{-1} \partial_\mu g) + \epsilon^* \partial_\mu (g^{-1} \partial_\mu g)^*].$$

In order that the sum of the  $\epsilon$  and  $\epsilon^*$  term vanish for arbitrary  $\epsilon$  it is necessary that the condition

$$\partial_\mu (g^{-1} \partial_\mu g) = 0$$

should be fulfilled as announced.

### Examples

(a)  $G = O(N)$ ; the star automorphism is the identity. The element  $g_0$  satisfies  $g_0^2 = \lambda \mathbb{1}$ ; as  $g_0$  is an orthogonal matrix,  $\lambda$  is restricted to be  $+1$  or  $-1$ .

(i)  $\lambda = +1$ . We can take without loss of generality,  $g_0$  diagonal. It possesses  $p$  eigenvalues equal to  $+1$  and  $(N-p)$  equal to  $-1$ . The subgroup  $H$  which leaves  $g_0$  invariant is the direct product  $O(p) \times O(N-p)$ . The special cases  $p = 1$  or  $(N-1)$  reduces to the usual  $O(N)$   $\sigma$ -model.

(ii)  $\lambda = -1$ , which implies  $N$  even:  $N = 2M$ ; again without loss of generality we can take

$$g_0 = \left[ \begin{array}{c|c} 0 & I \\ \hline -I & 0 \end{array} \right]$$

and the subgroup  $H$  of orthogonal transformations which leaves  $g_0$  invariant can be identified to the unitary group  $U(M)$ .

(b)  $G = U(N)$ .

(i) The star automorphism is the identity and thus  $g_0^2 = \lambda \mathbb{1}$ , but we can absorb the  $\lambda$  factor, which is necessarily of unit modulus, into a redefinition of  $g_0$ . Again the eigenvalues of  $g_0$  are plus or minus one and if there are  $p$  eigenvalues equal to plus one, the subgroup  $H$  is  $U(p) \times U(N-p)$ . For  $p = 1$  or  $N-1$  we recover the  $CP^{N-1}$   $\sigma$ -models introduced by Eichenherr [3].

(ii) The star automorphism is non-trivial; if we take complex conjugation for the star operation, the condition  $g_0 g_0^* = \lambda \mathbb{1}$  implies  $\lambda = \pm 1$ .

If  $\lambda = +1$ ,  $g_0$  can be taken to be the unit matrix (through an orthogonal unitary transformation followed by a diagonal unitary transformation). The subgroup  $H$  consists of unitary elements constrained to satisfy  $u^T u = \mathbb{1}$ ; it is thus the orthogonal subgroup  $O(N)$  of  $U(N)$ .

If  $\lambda = -1$  (this is possible only when  $N$  is even),  $g_0$  can be reduced to the form

$$g_0 = \left[ \begin{array}{c|c} 0 & I \\ \hline -I & 0 \end{array} \right]$$

and the subgroup  $H$  defined as

$$\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} u = u^* \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix},$$

is the symplectic subgroup  $\mathrm{Sp}(N)$  of  $\mathrm{U}(N)$ .

Though all these models have classically an infinite number of non-local conservation laws, it is not known whether this property holds for the quantum theory (apart from the simple  $\mathrm{O}(N)$   $\sigma$ -model [7]). It is not even known in general what would be the physical consequences of such conservation laws, except again for the  $\mathrm{O}(N)$   $\sigma$ -model for which Lüscher has shown [7] that they imply the absence of particle production and the factorisation of the  $S$ -matrix.

### 3. Quantization

We will first consider the models which seem to have the most interesting properties namely the models  $\mathrm{O}(N)/\mathrm{O}(p) \times \mathrm{O}(N-p)$  and  $\mathrm{U}(N)/\mathrm{U}(p) \times \mathrm{U}(N-p)$ ; we shall devote sect. 5 to the general case. Let us begin with the unitary model, the orthogonal one being very similar. In order to parametrize the elements of  $G/H$  it is convenient to introduce a set of projectors  $P(x)$  of rank  $p$  such that  $g(x) = 2P(x) - 1$ . In terms of  $P(x)$ , within a trivial rescaling, the action is

$$A = \frac{1}{2} \int dx \, \mathrm{tr} [\partial_\mu P(x) \partial_\mu P(x)],$$

with the constraints  $P(x) = P^+(x)$ ,  $P^2(x) = P(x)$ ,  $\mathrm{tr} P(x) = p$ . The classical minima of  $A$  consist of constant  $P(x) = P_0$  and a first method of quantization is to expand around one of the minima. This can be done with the parametrization

$$P(x) = \begin{bmatrix} \frac{1}{2} + (\frac{1}{4} - CC^+)^{1/2} & C \\ C^+ & \frac{1}{2} - (\frac{1}{4} - C^+C)^{1/2} \end{bmatrix},$$

in which  $C(x)$  is an arbitrary  $p \times (N-p)$  rectangular matrix and the square roots are defined by Taylor expanding and  $(\frac{1}{4} - CC^+)^{1/2} = \frac{1}{2} + \mathrm{O}(CC^+)$ . The quantum fields are the  $p \times (N-p)$  elements of  $C(x)$ .

An alternative (and more convenient in view of the large- $N$  limit) method is the parametrization through an orthonormal basis of the eigenspace into which  $P(x)$  projects:

$$P_{\alpha\beta}(x) = \sum_{i=1}^p \varphi_\alpha^i(x) \varphi_\beta^i(x), \quad \alpha, \beta = 1, \dots, N, \quad (7)$$

in which the  $p$  vectors  $\varphi_i(x)$  satisfy the orthonormality conditions

$$\varphi^{i*} \cdot \varphi^j = \sum_{\alpha=1}^N \varphi_\alpha^{i*} \varphi_\alpha^j = \delta_{ij}. \quad (8)$$

The equations (7), (8) define the set of vectors  $\varphi_i^i(x)$  only up to a  $U(p)$  unitary transformation at each point  $x$ . Therefore the action written in terms of the fields  $\varphi^i(x)$  will possess a local  $U(p)$  gauge invariance in addition to the  $U(N)$  global symmetry. Note that this procedure breaks the  $p \leftrightarrow (N-p)$  symmetry of the problem, but it will be seen below that the gauge invariant quantities do remain symmetric. In terms of these  $\varphi_i$  fields the action becomes

$$A = \int dx \{ \partial_\mu \varphi^{*i} \partial_\mu \varphi^i + \frac{1}{2} [ (\varphi^i \partial_\mu \varphi^{*i}) \cdot (\varphi^i \partial_\mu \varphi^{*i}) + \text{C.C.} ] \}. \quad (9)$$

This action being gauge invariant, the quantization requires that we fix the gauge. In order to avoid the appearance of Faddeev–Popov ghosts, we shall choose the analog of the unitary gauge of Higgs models. For that purpose let us decompose the vectors  $\varphi^i$  into upper and lower components

$$\varphi^i = \begin{pmatrix} U_k^i \\ V_\alpha^i \end{pmatrix} \quad k = 1, \dots, p, \quad \alpha = p+1, \dots, n. \quad (10)$$

The gauge freedom corresponds to the choice of an hermitian  $p \times p$  matrix which we can fix by the matrix requirement

$$U = U^+$$

in which

$$(U)_{ij} \equiv U_i^j. \quad (11)$$

The orthonormality conditions of the basis  $\varphi^i$  now read

$$(U^2)_{ij} = \delta_{ij} - V^{*i} \cdot V^j, \quad (12)$$

and the  $p \times (N-p)$  complex components of the vectors  $V^i$  are the independent field components.

There are measure induced terms which arise from the parametrization of the constraints in terms of the  $V^i$ , but we will adopt dimensional regularization in which these measure terms drop out [8].

In terms of the  $V^i$  the quantum action is now an infinite power series:

$$A = \frac{\mu^{d-2}}{t} \int d^d x \{ \partial_\mu V^{i*} \partial_\mu V^i + \frac{1}{2} [ (\partial_\mu V^{i*} \cdot V^j) (\partial_\mu V^{j*} \cdot V^i) + (V^{i*} \cdot \partial_\mu V^j) (V^{j*} \cdot \partial_\mu V^i) ] + O(V^6) \}, \quad (13)$$

in which the renormalization mass scale  $\mu$  has been introduced to make the coupling constant  $t$  dimensionless.

This theory is renormalizable by power counting in two dimensions [9]. If we limit ourselves to the calculation of the correlation functions of gauge invariant



been used. The results lead to the renormalization group functions

$$\beta(t) = \frac{\epsilon t}{1 + t \partial \log Z / \partial t}, \quad \epsilon = d - 2,$$

$$\gamma(t) = \beta(t) \frac{\partial}{\partial t} \log (Z_2 / Z_1);$$

we thus found in  $(2 + \epsilon)$  dimensions

$$\beta(t) = \epsilon t - N t^2 - 2[p(N - p) + 1]t^3 + O(t^4), \quad (15a)$$

$$\gamma(t) = -N t + O(t^3), \quad (15b)$$

for the unitary case, and

$$\beta(t) = \epsilon t - (N - 2)t^2 - [2p(N - p) - N]t^3 + O(t^4), \quad (16a)$$

$$\gamma(t) = -N t + O(t^3), \quad (16b)$$

for the orthogonal case.

### Remarks

- (i) A rescaling of the coupling constant  $t \rightarrow t/2\pi$  has been made.
- (ii) The symmetry  $p \leftrightarrow (N - p)$  is manifest.
- (iii) The orthogonal case when  $p = 1$  reduces to the usual  $O(N)$   $\sigma$ -model. Notice that the  $\gamma$  function corresponds to a composite operator in the  $O(N)$  language. It has been calculated in ref. [10] under the name  $\zeta_2$ .
- (iv) The unitary  $p = 1$  case ( $CP^{N-1}$  models) checks with the calculations of ref. [11].
- (v) The unitary  $p = 1$ ,  $N = 2$  model ( $CP^1$ ) reduces to  $O(3)$  up to a factor two in the definition of the coupling constant.

These models are asymptotically free in two dimensions; it has also been verified in ref. [12] for the chiral models  $G \times G$  in which  $G$  is either  $SU(N)$  or  $O(N)$ :

$$\beta(t) = -\frac{1}{4}(N^2 - 1)t^2 - \frac{1}{32}(N^2 - 1)^2 t^3 + O(t^4), \quad G = SU(N),$$

$$\beta(t) = -\frac{1}{4}N(N - 2)t^2 - \frac{1}{32}(N(N - 2))^2 t^3 + O(t^4), \quad G = O(N).$$

This asymptotic freedom is of course expected since a phase transition at zero temperature should take place for all non-abelian cases, as suggested by Mermin-Wagner theorem [13]. Consequently the origin in coupling constant space should be an infrared unstable fixed point, since in a massive theory decreasing the momenta reveals the presence of a non-zero mass term. In a one coupling constant theory infrared instability is synonymous of ultraviolet stability. In sect. 5 a proof of this statement will be given for symmetric spaces. We do not know if there are quantum models with a single coupling constant which do not belong to this class. Above

two dimensions a phase transition takes place, with a critical temperature of order  $(d-2)$ , and a correlation length exponent

$$\frac{1}{\nu} = -\beta'(t_c) = \epsilon + \frac{B}{A} \epsilon^2 + O(\epsilon^3),$$

if we write

$$\beta(t) = \epsilon t - At^2 - Bt^3 + O(t^4).$$

The anomalous dimension of the composite operator bilinear in  $\mathbf{V}$  and  $\mathbf{V}^*$  is given by

$$\gamma(t_c) = -\frac{N}{A} \left( \epsilon - \frac{B}{A^2} \epsilon^2 \right) + O(\epsilon^3).$$

We note in view of the possible applications suggested by Wegner [14] of these models, in the limiting case  $p = N = 0$ , to the mobility edge problem, that in this limit  $\gamma$  vanishes to all orders in  $\epsilon$ . Within this low-temperature expansion it is difficult to study the high temperature symmetric phase which extends in two dimensions up to  $t = 0$ . For these gauge symmetric generalized  $\sigma$ -models, not only the global symmetry is unbroken, but in addition in the high temperature phase the quanta corresponding to the constituent fields  $\mathbf{V}^i(x)$  are permanently confined by long range forces. Though this seems difficult to verify from perturbation theory it can be explicitly checked within the  $1/N$  expansion [4] which allows to go into the symmetric phase.

## 5. Asymptotic freedom of general symmetric spaces

We are going to consider the following quantum models: let  $|0\rangle$  be a fixed vector on which acts an irreducible representation of the compact gauge group  $G$ , and let  $H$  be the little group of this vector. The action of the group  $G$  on this vector generates cosets of  $G/H$ . The model is defined by a classical action density

$$A(x) = \langle \partial_\mu \varphi | \partial_\mu \varphi \rangle \quad (17)$$

in which

$$|\varphi(x)\rangle = D(g(x))|0\rangle; \quad (18)$$

$D(g)$  is an orthogonal representation of  $G$ .

This action has a global invariance under an arbitrary transformation of  $G$   $D(g(x)) \rightarrow D(g')D(g(x))$ . It has a local gauge invariance under  $D(g(x)) \rightarrow D(g(x))D(h(x))$  in which  $h(x)$  is an arbitrary element of  $H$ . This gauge invariance may be used to choose  $g(x)$  of the form

$$D(g(x)) = \exp \sum_1^l t^\alpha \xi^\alpha(x), \quad (19)$$

in which the  $t^\alpha$ ,  $\alpha = 1, \dots, l$  are the elements of Lie algebra of  $G$  not contained in the Lie algebra of  $H$ ; consequently

$$\sum_1^l x_\alpha t^\alpha |0\rangle \Leftrightarrow x^\alpha = 0, \quad \alpha = 1, \dots, l.$$

We have chosen to normalize the generators as real antisymmetric matrices, orthogonal under the trace, by

$$\frac{1}{n} \sum_1^n t^\alpha t^\alpha |0\rangle = -|0\rangle, \quad (20)$$

in which  $n$  is the number of generators of  $G$  (there are  $(n-l)$  generators of  $H$ ). Up to now we have considered general coset spaces  $G/H$  but we will have to impose the restrictions (i) that the quantum theory remains with only one coupling constant, and (ii) that the vector  $|\varphi(x)\rangle$  will be multiplicatively renormalized. This implies in particular that we have

$$\langle 0 | t^\alpha t^\beta | 0 \rangle = -a_\alpha \delta^{\alpha\beta}, \quad (21)$$

with

$$a_\alpha = 1 \quad \text{for} \quad \alpha \leq l, \quad (21)$$

$$a_\alpha = 0 \quad \text{by definition for} \quad l < \alpha \leq n. \quad (22)$$

The normalization of the vector  $|0\rangle$  is therefore fixed by (20) to be

$$\langle 0 | 0 \rangle = l/n. \quad (23)$$

We do not know if these conditions are met by some homogeneous non-symmetric spaces. However they hold for symmetric spaces. Indeed in the construction given in sect. 2 the fixed vector  $|0\rangle$  was the group element  $g_0$  itself which satisfied

$$g_0 g_0^* = \lambda \mathbb{1}$$

and the action of  $G$  on this vector was given as

$$(D(g)|0\rangle)_{ij} = \varphi_{ij} = (g^{-1*} g_0 g)_{ij}. \quad (24)$$

Let  $\tau_\alpha$  be the generators of the representation in which  $g_0$  and  $u$  are written. Eq. (24) in which  $u$  is chosen to be near the identity implies

$$(t^\alpha |0\rangle)_{ij} = (g_0 \tau^\alpha - \tau^{\alpha*} g_0)_{ij}. \quad (25)$$

Thus

$$\begin{aligned} \langle 0 | t^\beta t^\alpha | 0 \rangle &= \text{tr} (g_0^* \tau^\beta - \tau^{\beta*} g_0^*) (g_0 \tau^\alpha - \tau^{\alpha*} g_0) \\ &= \text{tr} [g_0 \tau^\alpha g_0^* \tau^{*\beta} + g_0 \tau^\beta g_0^* \tau^{*\alpha} - \lambda (\tau^\alpha \tau^\beta + \tau^{\alpha*} \tau^{\beta*})]. \end{aligned} \quad (26)$$

The Lie algebra may be decomposed as a sum of matrices which are even and odd under the automorphism

$$x \rightarrow \bar{x} = g_0^{-1} x^* g_0 \quad (27)$$

by the formula

$$\tau = \tau^+ + \tau^- \quad (28)$$

in which

$$\tau^\pm = \frac{1}{2}(\tau \pm g_0^{-1} \tau^* g_0). \quad (29)$$

Obviously one has

$$\bar{\tau}^\pm = \pm \tau^\pm, \quad (30)$$

or equivalently

$$\tau^{\pm*} g_0 = \pm g_0 \tau^\pm. \quad (31)$$

The elements of the form  $\tau^+$  are thus those of the Lie algebra  $L_+$  of  $H$ :

$$\text{Tr}(g_0 \tau^\alpha g_0^* \tau^{*\beta}) = \begin{cases} \lambda \text{Tr} \tau^{*\alpha} \tau^{*\beta}, & l < \alpha \leq n, \\ -\lambda \text{Tr} \tau^{*\alpha} \tau^{*\beta}, & \alpha \leq l. \end{cases} \quad (32)$$

Using (26) and (32) we obtain

$$\langle 0 | t^\beta t^\alpha | 0 \rangle = \begin{cases} 0 & \text{if } \alpha \text{ or } \beta > l, \\ -2\lambda \text{Tr}(\tau^\alpha \tau^\beta + \tau^* \tau^{*\beta}) & \alpha, \beta \leq l, \end{cases} \quad (33)$$

and if we assume that the generators of  $G$  are orthonormal by the trace, we do obtain the property which we required.

The quantum theory will now be defined by the action density

$$\begin{aligned} 2tZ_1 A(x) &= \langle 0 | \partial_\mu \exp \left\{ -\sum_1^l t_\alpha \xi_\alpha(x) \right\} \partial_\mu \exp \left\{ -\sum_1^l t_\alpha \xi_\alpha(x) \right\} | 0 \rangle \\ &\quad - 2Z_1 Z_2 m^2 \langle 0 | \exp \left\{ \sum_1^l t_\alpha \xi_\alpha(x) \right\} | 0 \rangle \end{aligned} \quad (34)$$

in which we have included a soft-mass term multiplicatively renormalizable.

Dimensional regularization will be used and all integration measure terms can be ignored. In order to write the vertices contained in (34) which are required for a one-loop calculation, we need the following identities:

$$[t^\alpha, t^\beta] = \sum_{\gamma=1}^n f^{\alpha\beta\gamma} t^\gamma \quad (35)$$

in which the structure constants  $f^{\alpha\beta\gamma}$  are real and completely antisymmetric;

$$\sum_{\beta, \gamma=1}^n f^{\alpha\beta\gamma} f^{\alpha'\beta\gamma} = c \delta^{\alpha\alpha'}, \quad (c \text{ is real and positive}); \quad (36)$$

$$\langle 0 | t^\alpha t^\beta t^\gamma | 0 \rangle = -\frac{1}{2} f^{\alpha\beta\gamma} (a_\alpha - a_\beta + a_\gamma), \quad (37)$$

which follows from  $\langle 0 | t^\alpha t^\beta t^\gamma + t^\gamma t^\beta t^\alpha | 0 \rangle = 0$  (the  $a_\alpha$ 's are defined by eqs. (21), (22)). The index sum in one-loop diagrams will rely upon the equations

$$\sum_{\alpha, \beta=1}^l \langle 0 | t^\alpha t^\alpha t^\beta t^\beta | 0 \rangle = ln, \quad (38a)$$

$$\sum_{\alpha, \beta=1}^l \langle 0 | t^\alpha t^\beta t^\alpha t^\beta | 0 \rangle = l(n - \frac{1}{2}c), \quad (38b)$$

$$\sum_{\alpha, \beta=1}^l \langle 0 | t^\alpha t^\beta t^\beta t^\alpha | 0 \rangle = l(n - \frac{1}{2}c) + \frac{1}{2} \sum_{\alpha; \beta; \gamma=1}^l (f^{\alpha\beta\gamma})^2. \quad (38c)$$

This gives the one-loop action

$$\begin{aligned} 2iZ_1 A(x) = & -\frac{l}{n} m^2 Z_1 Z_2 + \frac{1}{2} \sum_{\alpha=1}^l [(\partial_\mu \xi_\alpha)^2 + m^2 Z_1 Z_2 (\xi_\alpha)^2] \\ & + \sum_{\alpha, \beta, \gamma, \delta=1}^l \frac{1}{24} \langle 0 | t_\alpha t_\beta t_\gamma t_\delta | 0 \rangle \sum_{q_1+q_2+q_3+q_4=0} \xi_\alpha(q_1) \xi_\beta(q_2) \xi_\gamma(q_3) \xi_\delta(q_4) \\ & \times [3(q_1+q_2)^2 - 2(q_1^2 + q_4^2) - m^2] + O(\xi^6). \end{aligned} \quad (39)$$

We have calculated the vacuum energy  $W(m^2)$  and demanded that it remains finite after minimal subtraction

$$\begin{aligned} W = & \frac{m^2 l}{nt} Z_2 - \frac{l}{2} \int d^d q \ln \left( 1 + \frac{m^2}{q^2} Z_1 Z_2 \right) \\ & - \frac{t}{8} m^2 \left[ l(n-c) + \frac{1}{2} \sum_{\alpha, \beta, \gamma=1}^l (f_{\alpha\beta\gamma})^2 \right] \times \left[ \int \frac{d^d q}{q^2 + m^2} \right]^2; \end{aligned} \quad (40)$$

this leads to

$$Z_1 = 1 + \frac{1}{2l} \left\{ lc - \frac{1}{2} \sum_{\alpha, \beta, \gamma=1}^l (f_{\alpha\beta\gamma})^2 \right\} \frac{t}{\epsilon} + O(t^2), \quad (41)$$

$$Z_2 = 1 - \frac{nt}{2\epsilon} + \frac{t^2}{\epsilon^2} \frac{n}{8l} \left\{ l(n-c) + \frac{1}{2} \sum_{\alpha, \beta, \gamma=1}^l (f_{\alpha\beta\gamma})^2 \right\} + O(t^3). \quad (42)$$

The renormalization group functions

$$\beta(t) = \frac{\epsilon t}{1 + t \frac{\partial}{\partial t} \log Z_1}, \quad \gamma(t) = \beta(t) \frac{\partial \log Z_2}{\partial t},$$

are then given as

$$\beta(t) = - \left( \frac{C}{2} - \frac{1}{4l} \sum_{\alpha, \beta, \gamma=1}^l (f_{\alpha\beta\gamma})^2 \right) t^2 + O(t^3), \quad (43)$$

$$\gamma(t) = -\frac{tn}{2} + O(t^3). \quad (44)$$

Asymptotic freedom follows since from the definition (36)

$$lC = \sum_{\alpha=1}^l \sum_{\beta,\gamma=1}^n (f^{\alpha\beta\gamma})^2 > \sum_{\alpha,\beta,\gamma=1}^l (f^{\alpha\beta\gamma})^2,$$

and the coefficient of  $t^2$  in  $\beta(t)$  is thus negative. Note that for symmetric spaces  $f^{\alpha\beta\gamma}$  vanishes if  $\alpha, \beta, \gamma$  are all smaller than  $l$ . This follows from the decomposition into odd and even elements of the Lie algebra with respect to the operation  $x \rightarrow \bar{x}$ . The commutator of two odd elements is even. The formulae which are here contain all the previous one-loop results (but the normalization of the coupling constant has to be adjusted for numerical comparisons).

As a final remark we note that the method followed here is entirely equivalent to the one in which ones starts with the action density  $A(x) = \text{Tr} (g^{-1} D_\mu g)^2$  in which  $g(x) \in G$  and  $D_\mu = \partial_\mu + A_\mu$ ,  $A_\mu$  belonging to the Lie algebra of a subgroup H.

## 6. Large- $N$ limit

We do not know how to study in general the large- $N$  limit. The method that we shall use, which is a saddle-point expansion, is applicable only when the ratio of the number of generators of the subgroup H to the number of generators of the whole group G approaches one when  $N$  goes to infinity. Let us consider for definiteness the orthogonal case  $O(N)/O(p) \times O(N-p)$ . It is useful to notice that these models are limits of infinite bare charge of gauge fields coupled to scalar fields: indeed let us introduce a gauge field  $A_\mu^{ij}$  associated to the gauge group  $O(p)$  and the action density

$$A(x) = \frac{1}{4e_0^2} \text{tr} F_{\mu\nu}^2 + (D_\mu^{ij} \phi^j)^2, \quad (45)$$

with the non-linear constraints  $\phi^i, \phi^j = \delta^{ij}$ ,  $i, j = 1, \dots, p$ .

The charged scalar fields  $\phi^i$  are the same as above; each  $\phi^i$  has  $N$  components, and the covariant derivative  $D_\mu$  is defined as

$$D_\mu^{ij} = \delta_{ij} \partial_\mu + A_\mu^{ij}, \quad (46)$$

and  $F_{\mu\nu}$  is related in the usual way to  $A_\mu^{ij}$ .

In the formal limit  $e_0 \rightarrow \infty$ , this action does not involve any derivative of the gauge fields which can then be integrated out leading to the action on the remaining scalar fields

$$A(x) = \partial_\mu \phi^i \partial_\mu \phi^i - (\phi^j \cdot \partial_\mu \phi^i)^2, \quad (47)$$

with the same non-linear constraints on the  $\phi$ 's. It should though be noted that for any finite  $e_0$  the gauge group is simply  $O(p)$ , whereas the limiting model has an additional  $p \leftrightarrow (N-p)$  symmetry.

In order to study the large- $N$  (fixed  $p$ ) limit it is more convenient to keep the gauge field  $A_\mu$  and to impose the non-linear constraints on the scalar fields through Lagrange multipliers. We thus consider the action density

$$A(x) = \frac{1}{2t} \{ \partial_\mu \phi^i \partial_\mu \phi^i + 2A_\mu^{ij} (\phi^i \cdot \partial_\mu \phi^j) + A_\mu^{ij} A_\mu^{ij} \} \\ + \lambda \left( \sum_i \phi^i{}^2 - p \right) + \mu_{ij} \left[ \left( \phi^i \phi^j - \frac{\delta^{ij}}{p} \sum_{k=1}^p (\phi^k)^2 \right) \right], \quad (48)$$

with  $\text{tr } \mu = 0$ .

We split as before  $\phi^i$  into upper and lower components

$$u_\alpha^i = \phi_\alpha^i, \quad \alpha = 1, \dots, p, \\ v_\alpha^i = \phi_\alpha^i, \quad \alpha = p+1, \dots, N, \quad (49)$$

and perform explicitly the integrations over the  $v_\alpha^i$ . The resulting effective action density is

$$A(x) = \frac{1}{2t} \left\{ (D_\mu^{ij} u^j) + \lambda \left( \sum_i u_i^2 p \right) + \mu_{ij} \left( u^i u^j - \frac{\delta^{ij}}{p} \sum_k (u^k)^2 \right) \right\} \\ + \frac{1}{2} (N-p) \text{tr} \log [\lambda \delta_{ij} + \mu_{ij} - D_\mu^{ik} D_\mu^{kj}] - \frac{h}{2t} \sum_i u_i^2. \quad (50)$$

The last term is an explicit symmetry breaking term which allows to describe the nature of the phases. The dependence upon  $N$  is now explicit since the vectors and matrices have an  $N$ -independent number of components.

If we take  $t$  of order  $1/N$  the action is proportional to  $N$  and therefore the steepest descent method may be used to generate a systematic  $1/N$  expansion [8]. The theory is supposed to be regularized by an unspecified ultraviolet cut-off\*. The saddle-point equations read:

$$A_\mu^{ij} = 0, \\ \lambda u_i + \mu_{ij} u_j = h u_i, \\ \frac{1}{t} \left( p - \sum_i u_i^2 \right) = (N-p) \text{tr} \left[ \frac{1}{\lambda \delta_{ij} + \mu_{ij} - \nabla^2 \delta_{ij}} \right], \\ \frac{1}{t} \left( u^i u^j - \frac{\delta^{ij}}{p} \sum_k u_k^2 \right) = -(N-p) \left\{ \left( \frac{1}{(\lambda - \nabla^2) 1 + \mu^2} \right)_{ij}, \right. \\ \left. - \frac{\delta_{ij}}{p} \text{tr} \left( \frac{1}{(\lambda - \nabla^2) 1 + \mu} \right) \right\}. \quad (51)$$

\* The renormalizability of the  $1/N$  expansion of the  $\text{CP}^N$  models has been systematically studied in refs. [15].

There are two different possible solutions.

### 6.1. High temperature

We take  $\mu_{ij} = 0$ ,  $u_i = 0$  and  $\lambda_0$  is given by the equation

$$\frac{1}{t} = (N-p) \int^{\Lambda} \frac{d^d q}{\lambda_0 + q^2}. \quad (52)$$

This phase exists for all values of  $t$  larger than the critical temperature  $t_c$  at which  $\lambda_0$  vanishes:

$$\frac{1}{t_c} = (N-p) \int^{\Lambda} \frac{d^d q}{q^2}.$$

In two dimensions  $\lambda_0$  vanishes only when  $t$  goes to zero and there is no low-temperature phase.

We see that an expectation value  $\lambda_0$  for  $\lambda$  contributes to a mass term for the fields  $u^i$ . This mass is non-zero even in the  $h = 0$  limit and is exponentially small for  $t$  small in two dimensions. For  $h = 0$  the global  $O(N)$  symmetry is unbroken.

In the special case of the non-linear  $\sigma$ -model ( $p = 1$ ) we read the spectrum of the model out of this effective action; it consists of a single  $N$ -plet of massive particles. For  $p > 1$  there are still gauge fields, which as we shall see, remain massless since the gauge invariance is unbroken and the situation needs a careful analysis. Let us calculate the propagators which enter in the  $1/N$  expansion by taking the quadratic fluctuations of the action around the saddle point ( $\lambda = \lambda_c + \bar{\lambda}$ ):

$$\begin{aligned} A^{\text{QU}} = & \frac{1}{2t} \sum_i [(\partial_\mu u_i)^2 + \lambda_c u_i^2] + \frac{1}{2}(N-p) \text{tr} \left[ \frac{1}{\lambda_c - \nabla^2} A_\mu^{ik} A_\mu^{ki} \right] \\ & - \frac{1}{4}(N-p) \text{tr} \left\{ \frac{1}{\lambda_c - \nabla^2} (\delta_{ij} \bar{\lambda} + \mu_{ij} - \partial_\mu A_\mu^{ij} - A_\mu^{ij} \partial_\mu) \right. \\ & \left. \times \frac{1}{\lambda_c - \nabla^2} (\delta_{ij} \bar{\lambda} + \mu_{ij} + \partial_\mu A_\mu^{ij} + A_\mu^{ij} \partial_\mu) \right\}. \end{aligned} \quad (53)$$

One sees as expected the massive propagator for  $u_i$ , and for the gauge field the quadratic part may be written as

$$\frac{1}{2}(N-p) A_\mu^{ij}(q) A_\nu^{ji}(-q) \left\{ \delta_{\mu\nu} \int \frac{d^d k}{k^2 + \lambda_c} - \frac{1}{2} \int \frac{(2k-q)_\mu (2k-q)_\nu d^d k}{(k^2 + \lambda_c)[(q-k)^2 + \lambda_c]} \right\}, \quad (54)$$

and in any reasonable regularization scheme (for instance dimensional) one can

check easily that

$$q_\mu \int \frac{(2k-q)_\mu (2k-q)_\nu d^d k}{(k^2 + \lambda_c)[(q-k)^2 + \lambda_c]} = 2q_\nu \int \frac{d^d k}{k^2 + \lambda_c}.$$

Therefore the quadratic part in the gauge field is purely transverse and the field is thus massless. This shows the existence of a long range Coulomb force which couples the  $O(p)$  charged particles. Thus in two dimensions as in QED or in the  $CP^N$  models [4] the charged particles are confined. The resulting spectrum may be obtained approximatively in the finite  $tN$ , large  $N$  limit by a non-relativistic approximation: in this regime there is a weak long range force (of order  $1/N$ ) between the massive confined constituents. In contradistinction, with the usual  $O(N)$   $\sigma$ -model, both the perturbative and  $1/N$  expansions are affected by infrared instabilities. The physical spectrum would probably emerge more naturally from the limit in which  $p$  and  $N-p$  go to infinity at the same rate; however this limit involves a summation of planar Feynman diagrams.

## 6.2. Low temperature

The saddle-point solution is still  $A_\mu^{ij} = 0$ ,  $\mu_{ij} = 0$  but we are now in the phase in which  $\lambda_0 = h$ , and  $u_{i0}$  is non-zero even when  $h$  vanishes. The saddle-point equations are then

$$u_{i0}u_{j0} = \sigma^2 \delta_{ij},$$

$$1 - \sigma^2 = t(N-p) \int \frac{d^d q}{q^2 + h}.$$

In the limit  $h = 0$  this phase exists only above two dimensions for  $t$  below the critical temperature  $t_c$ . Quantum fluctuations around this saddle-point equations show that the gauge field acquires a mass since the quadratic part of the action involving the gauge field is

$$\frac{1}{2t} A_\mu^{ij}(q) A_\nu^{ji}(-q) \left\{ \delta_{\mu\nu} \sigma^2 + t(N-p) \left[ \delta_{\mu\nu} \int d^d k \frac{1}{k^2 + h} - \frac{1}{2} \int \frac{(2k-q)_\mu (2k-q)_\nu}{(k^2 + h)[(q-k)^2 + h]} d^d k \right] \right\}.$$

The second term is transverse, but when  $\sigma$  is non-zero the gauge field is massive: a Higgs phenomenon has taken place. The counting of degrees of freedom may be described as follows:

(i) in the high-temperature phase there are  $Np$  field components and  $\frac{1}{2}p(p+1)$  constraints  $\varphi^i \varphi^j = \delta_{ij}$ ;

(ii) in the low-temperature phase there are  $\frac{1}{2}p(p-1)$  massive gauge fields coupled to  $p(N-p)$  Goldstone bosons which are the quantum fields of our low temperature expansion.

## References

- [1] V.L. Golo and A.M. Perelomov, Phys. Lett. 79B (1978) 112;  
J. Fröhlich, IHES preprint (1979);  
M. Dubois-Violette and Y. Georgelin, Phys. Lett. 82B (1979) 251.
- [2] E. Brezin, C. Itzykson, J. Zinn-Justin and J.B. Zuber, Phys. Lett. 82B (1979) 442.
- [3] H. Eichenherr, Nucl. Phys. B146 (1978) 215.
- [4] A. D'Adda, P.Di Vecchia and M. Lüscher, Nucl. Phys. B146 (1978) 63;  
E. Witten, Nucl. Phys. B149 (1979) 285.
- [5] S. Coleman, J. Wess and B. Zumino, Phys. Rev. 177 (1969) 2239.
- [6] H. Eichenherr and M. Forger, Nucl. Phys. B155 (1979) 381;  
R. Pisarski, Princeton Univ. preprint (1979).
- [7] M. Lüscher, Nucl. Phys. B135 (1978) 1.
- [8] E. Brezin and J. Zinn-Justin, Phys. Rev. B14 (1976) 3110.
- [9] E. Brezin, J. Zinn-Justin and J.C. Le Guillou, Phys. Rev. D14 (1976) 2615.
- [10] E. Brezin, J. Zinn-Justin and J.C. Le Guillou, Phys. Rev. D14 (1976) 4976.
- [11] S. Hikami, Progr. Theor. Phys. 62 (1979) 226.
- [12] A. McKane and M. Stone, Oxford Univ. preprint (1979).
- [13] N. Mermin and H. Wagner, Phys. Rev. Lett. 17 (1966) 1133.
- [14] F. Wegner, The mobility edge problem; continuous symmetry and a conjecture, Heidelberg Univ. preprint (1979).
- [15] I. Araf'eva, Steklov Inst. preprint;  
E. Nissimov and S. Pacheva, Sofia preprint.